NETWORKING

VPNs
Visual Guide

Tunnels through the internet —
private networks over public wires.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

The Tunneling Concept

Wrap private packets inside public ones — the internet sees only the tunnel.

- Encapsulation: the original packet becomes the PAYLOAD of a new outer packet.
- Encryption + authentication: confidentiality plus proof nobody tampered in transit.
- To your apps, the tunnel looks like a local network interface (tun0).

STEP 2/5

IPsec vs WireGuard vs OpenVPN

Three generations of tunneling.

IPsec

IETF standard, kernel-level. IKEv2 key exchange. Complex but ubiquitous — site-to-site default.

OpenVPN

TLS-based, user space. Very flexible, very configurable — and slower for it.

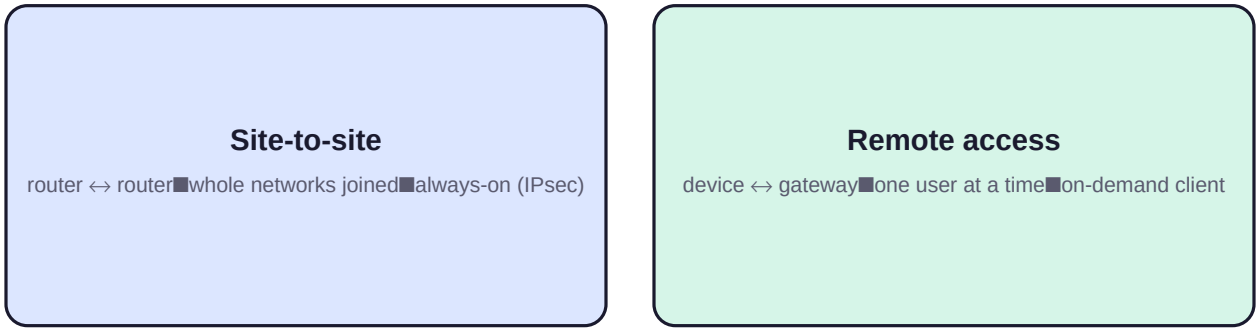
WireGuard

Modern, ~4k lines. ChaCha20/Noise. Roams across networks; connects in milliseconds.

New deployments default to WireGuard unless compliance or legacy gear demands IPsec.

STEP 3/5

Site-to-Site vs Remote Access



- Site-to-site: branch offices behave as one LAN — printers, file shares, internal DNS all reachable.
- Remote access: the traveling laptop joins the office net; split-tunnel keeps Netflix local, full-tunnel routes everything.
- Zero-trust trend: per-app authenticated access is replacing 'VPN = inside the castle' thinking.

STEP 4/5

What a VPN Packet Looks Like

Headers inside headers — each layer adds its envelope.

```
[ outer IP | ESP header | inner IP | TCP | app data | ESP trailer/auth ]

  ■■ routable ■■ ■■■■■ encrypted + authenticated ■■■■■
```

- ESP (Encapsulating Security Payload) provides the encryption and integrity in IPsec tunnel mode.
- An observer sees: tunnel endpoints, packet sizes, timing — NOT contents. Traffic analysis is still possible.
- MTU shrinks: the tunnel eats 40-80 bytes — clamp MSS or fragment to avoid black-hole stalls.

STEP 5/5

Encryption Overhead & Gotchas

Security costs something — budget for it.

- Throughput: encryption is cheap on modern CPUs (AES-NI); the bottleneck is usually the tunnel endpoint's bandwidth.
- Latency: +1 RTT-ish for crypto plus the detour via the VPN gateway — pick nearby exit points.
- DNS leaks: a VPN that doesn't tunnel DNS exposes every lookup — verify with a leak test.
- Kill switch: if the tunnel drops, block traffic rather than silently sending it in the clear.

A consumer VPN hides traffic from the local network/ISP — but the VPN provider sees it all. Trust is the real protocol.

STEP ✓I✓

Key Takeaways

VPNs — Remember

- Tunneling = private packets wrapped as payload inside public, encrypted packets.
- IPsec (site-to-site standard), OpenVPN (flexible TLS), WireGuard (modern default).
- Site-to-site joins networks; remote access joins individual devices.
- Observers still see endpoints, sizes, timing — encryption hides contents, not metadata.
- Watch MTU shrinkage, DNS leaks, and always run a kill switch.

Keep exploring

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